THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** the volume of water is changed along with the time, so the test has two changing things	Volume of water is a variable.
2	**(a)** both depend on hot weather	A third factor links them.
3	**(b)** compare the methods and repeat carefully	Differences in method may explain.
4	**(a)** more than one variable was changed	Only one variable should vary.
5	**(d)** amount of water	Water amount varies.
6	**(b)** rounding off numbers to look neat without noting it	Data must not be altered.
7	**(d)** revise the hypothesis	Data must be respected.
8	**(a)** growth of the plant	Growth is the dependent variable.
9	**(b)** Why do wet clothes dry faster on a windy day?	It looks for a cause.
10	**(a)** Does temperature affect how fast sugar dissolves?	Testable questions have measurable answers.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Two factors differ: light and water (1); We cannot tell which factor caused the difference (1).	1+1
12	It represents the Earth in small size (1); It helps us study things too large to observe wholly (1).	1+1
13	Example: plastic waste (1); Science develops recycling or biodegradable materials (1).	1+1
14	It shows what happens without the change (1); Example: unfertilised plants compared to fertilised ones (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use same size, same wetness (1); Same place, time and temperature (1); Measure drying time for each (1).	1+1+1
16	Curiosity: why do things fall (1); Observation: fall of apple and Moon's motion (1); Reasoning: same force acts on both (1).	1+1+1
17	A factor that can change (1); Temperature of water (1); Amount of sugar and stirring (1).	1+1+1
	OR: P vs Q tests light (1); P vs R tests water (1); Only one factor differs in each comparison (1).	1+1+1
18	Hand is unreliable (1); It depends on earlier experience (1); A thermometer gives exact readings (1).	1+1+1
	OR: New experiments contradict it (1); A better explanation covers more facts (1); Example: Earth at centre idea was given up (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hypothesis: dark cups cool faster (1); Same volume and temperature of water (1); Cups of different colour, same size and material (1); Measure temperature every 5 minutes (1); Plot results and conclude (1).	1+1+1+1+1
	OR: Two sets of same plants (1); One with music, one without (control) (1); Same water, soil, light (1); Measure height weekly (1); Compare and conclude (1).	1+1+1+1+1
20	Sum = 90 s (1); Average = 30 s (1); One swing = 1.5 s (1); Timing many swings reduces error in starting and stopping (1); Dividing gives the time for one swing (1).	1+1+1+1+1
	OR: Parallax error (1); Wrong zero mark (1); Using broken scale (1); Reading from the correct angle (1); Repeating and averaging (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** careful observation matters more than costly equipment alone	Simple tools with careful thinking can lead to discoveries.
2	**(b)** the plant grows with time	Rising line indicates growth.
3	**(d)** the half field not sprayed	The control is the unchanged group.
4	**(d)** it follows from the data collected under fair conditions	Validity depends on evidence.
5	**(d)** they can measure precisely and consistently	Precision and consistency.
6	**(c)** repeated measurements by different people	Repeated, independent evidence is strong.
7	**(c)** The rod is 25 cm long	It involves a number.
8	**(c)** verification	Others can verify.
9	**(c)** no single instrument can measure everything	Each measures a different quantity.
10	**(d)** The solution is blue	Colour is a quality.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	One case may be a coincidence (1); More cases are needed to see a pattern (1).	1+1
12	It is a single case (1); He may have recovered naturally; trials are needed (1).	1+1
13	Average = 39 / 3 = 13 s (1); It reduces the effect of random errors (1).	1+1
14	A very different value stands out (1); It can be examined and the cause found (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Two things may rise together (1); Example: ice-cream sales and sunburn (1); Both depend on a third factor, hot weather (1).	1+1+1
16	It can be tested with evidence (1); Scientific: does heat affect melting of wax (1); Non-scientific: which flower is lovelier (1).	1+1+1
17	Eyes see colour and shape (1); Nose notes smell (1); Magnifying lens shows the tiny parts (1).	1+1+1
	OR: Different measuring habits (1); Different conditions such as temperature (1); Errors in instruments or reading (1).	1+1+1
18	They used observation and reasoning (1); Their records influenced later knowledge (1); They inspire scientific attitude (1).	1+1+1
	OR: Honest data builds trust (1); Teams share work, such as large experiments (1); Different skills solve complex problems (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Define the claim clearly (1); Set up controlled tests with and without contact (1); Measure responses with instruments (1); Repeat by other groups (1); Accept only if supported repeatedly (1).	1+1+1+1+1
	OR: Advantage: faster communication (1); Advantage: better health care (1); Advantage: easier travel (1); Harms: pollution and waste (1); Science finds cleaner solutions (1).	1+1+1+1+1
20	Telescope revealed moons and stars (1); Microscope revealed cells (1); Earlier beliefs changed (1); New questions arose (1); New branches of science developed (1).	1+1+1+1+1
	OR: Science answers testable questions (1); Many questions remain unanswered (1); New discoveries pose new questions (1); Values and opinions are outside science (1); Knowledge keeps growing (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** It can have both benefits and harmful effects	Responsible use is important.
2	**(d)** question, hypothesis, test, result, conclusion, sharing	This is the logical sequence.
3	**(c)** new evidence can change them	They may be revised.
4	**(d)** cells and microbes	Microscopes revealed the unseen world of organisms.
5	**(d)** to represent things too large or small to handle, like the solar system	Models help to visualise.
6	**(c)** others can get the same result	Replication builds trust.
7	**(b)** measuring your height	Simple measurement needs no hypothesis.
8	**(a)** it can only answer questions that can be tested	Science studies testable questions.
9	**(a)** Time taken for ice to melt at different temperatures	Titles should state what is measured.
10	**(d)** to check safety and effect through tests	Trials ensure safety.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Others check methods and results (1); It catches errors and improves quality (1).	1+1
12	It is an opinion that cannot be tested (1); There is no measurable evidence (1).	1+1
13	It would be a guess rather than a result (1); Only what the evidence supports can be claimed (1).	1+1
14	A common system avoids confusion (1); Results can be compared everywhere (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A metal rod of same length after heating (1); If length stays the same (1); Repeated trials show no change (1).	1+1+1
16	Table organises readings (1); Graph shows pattern or trend (1); Both make comparison easy (1).	1+1+1
17	Measure mass before soaking (1); Measure mass after soaking (1); Difference is the water absorbed (1).	1+1+1
	OR: Water purifiers (1); Rain water harvesting (1); Drip irrigation to save water (1).	1+1+1
18	Check for errors (1); Repeat the measurement (1); Not delete it without reason (1).	1+1+1
	OR: Dirt can change results (1); Old chemicals can react (1); It prevents contamination (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Data: readings like 5 cm, 8 cm (1); Evidence: data that supports a claim (1); Conclusion: statement drawn from evidence (1); Example: plant heights in light (1); Conclusion: light helps growth (1).	1+1+1+1+1
	OR: Aryabhata: Aryabhatiya (1); Charaka: Charaka Samhita (1); Sushruta: Sushruta Samhita (1); Based on observation and reasoning (1); Passed on through written texts (1).	1+1+1+1+1
20	Question and observation (1); Hypothesis (1); Tests and data (1); Sharing and review (1); Repeated confirmation (1).	1+1+1+1+1
	OR: Change only light level (1); Control: same seeds, soil, water (1); Measure height daily with a ruler (1); Record in a table (1); Handle soil and tools safely (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.